Lewis on counterpart theory

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1. Modal realism

2. The theory

P1: $\forall x \forall y(Ixy \rightarrow Wy)$ (Nothing is in anything except a world)

P2: $\forall x \forall y \forall z((Ixy \wedge Ixz) \rightarrow y = z)$ (Nothing is in two worlds)

P3: $\forall x \forall y(Cxy \rightarrow \exists z Ixz)$ (Whatever is a counterpart is in a world)

P4: $\forall x \forall y(Cxy \rightarrow \exists z Iyz)$ (Whatever has a counterpart is in a world)

P5: $\forall x \forall y \forall z((Ixy \wedge Izy \wedge Cxz) \rightarrow x = z)$ (Nothing is a counterpart of anything else in its world)

P6: $\forall x \, \forall y(Ixy \rightarrow Cxx)$ (Anything in a world is a counterpart of itself)

P7: $\exists x(Wx \wedge \forall y(Iyx \leftrightarrow Ay))$ (Some world contains all and only actual things)

P8: $\exists x Ax$ (Something is actual)

3. Puzzles about how the two languages relate to English

4. Rejected extra axioms

A1.$\forall xyz(Cxy \wedge Cyz \rightarrow Cxz)$ (Counterparthood is transitive)

A2.$\forall xyz(Cxy \rightarrow Cyx)$ (Counterparthood is symmetric)

A3.$\forall x \forall y \forall z \forall w((Cxz \wedge Cyz \wedge Ixw \wedge Izw) \rightarrow x=y)$ (At most one counterpart per world)

A4. $\forall x \forall y \forall z \forall w((Cxy \wedge Cxz \wedge Iyw \wedge Izw) \rightarrow y=z)$ (Worldmates don't share counterparts)

A5.$\forall x \forall v \forall w((Ixv \wedge Ww) \rightarrow \exists y(Cxy \wedge Iyw))$ (Everything is the counterpart of something in each world)

A6.$\forall x \forall v \forall w((Ixv \wedge Ww) \rightarrow \exists y(Cyx \wedge Iyw))$ (At least one counterpart per world)

5. The translation scheme

6. Extension to names

Lewis advocates treating names as definite descriptions that introduce scope ambiguities. If we don't think they introduce scope ambiguities (either because we are "mandatory wide scope descriptivists" or because we are not descriptivists at all), we can just treat them exactly as Lewis treats variables.

- This will however introduce new surprises if we have names for things that aren't in @. For example, if a is a name for a dragon that isn't in @, the translation of Dragon(a) ∧ ¬∃xDragon(x) will be true.
- Is there already a problem from the fact that translations of *open* theorems of first order logic, like Dragon(y) → ∃xDragon(x), are open non-theorems?

7. Surprises in nonmodal logic: failures of quantified LL

Failures of quantified LL

$$\forall x \forall y(x=y \rightarrow (\Box(x=x) \rightarrow \Box(x=y)))$$

translates as:

$\forall x(Ix@ \rightarrow \forall y(Iy@ \rightarrow (x=y \rightarrow$

$(\forall w \forall z((Ww \wedge Izw \wedge Czx) \rightarrow z=z) \rightarrow \forall w \forall z \forall u((Ww \wedge Izw \wedge Iuw \wedge Czx \wedge Cuy) \rightarrow z=u)))$

The translation is *not* a theorem, although it would be one if we added A3.

How does this play out in English? For example, sentences like 'There is a statue on the table that could have outlived each statue on the table' can be true (even when the 'each' scopes over the 'could'). Seems weird!

- It's worth noting that we *do* get LL-instances $\forall x \forall y(x=y \rightarrow (\Phi[x/z] \rightarrow \Phi[y/z]))$ when neither x nor y is free in Φ, and those where both x and y are free in Φ. Does this make things better?
- Lewis's response in the afterword: .

8. Restriction to qualitative predicates?

Even if we added A3 to the system, there is still a different kind of LL-worry

Consider ∃y□∀x(From(x, y) ↔ German(x)). The translation is:

(*) $\exists y(Iy@ \wedge \forall w \forall z((Ww \wedge Czy \wedge Izw) \rightarrow \forall x(Ixw \rightarrow (From(x,z) \leftrightarrow German(x))))$

If we combine this with ∀x(German(x) → Ix@), we get something very weird.

You might think you could deal with this by denying ∀x(German(x) → Ix@). But….

- If Germany is part of @, everything from Germany is in @; doesn't it follow that every German is in @?
- Suppose that for every country in @, there is a world where that country has two counterparts, such that some people in that world are from the one but not the other counterpart. Then there is no way for (*) to be true!?

A natural fix: say that the counterpart-theoretic translation scheme does not apply to just any old predicate, but only applies to *qualitative* predicates.

- Intuitive concept of qualitativeness: a qualitative property or relation is one that isn't "about"—doesn't "involve"; can be expressed without "implicitly referring to"; …—any *particular objects*. More on this next week.

9. The 'counterparts of tuples' alternative

10. Surprises in modal logic 1: necessitism

A controversial theorem: $\forall x \Box \exists y(y=x)$.

11. Surprises in modal logic 2: failures of K

The following schema plays a foundational role in most ordinary modal logics:

$$K: \Box(P \to Q) \to (\Box P \to \Box Q)$$

In Lewis's counterpart theory, not all universal closures of K-instances are theorems. For example, the following is not:

$$\forall x(\Box(Fx \to \exists yFy) \to (\Box Fx \to \Box\exists yFy))$$

Similarly, we have failures of the principle (which follows from K in the usual systems) that necessity commutes with conjunction:

$$CC: \Box(P \wedge Q) \leftrightarrow (\Box P \wedge \Box Q)$$

The right-to-left direction of this is such that all its universal closures are theorems; but the left-to-right direction isn't. For example, $\forall x\forall y(\Box(Fx \wedge Gy) \to \Box Fx)$ will be false if there is an x that has a non-F counterpart in some world, but is such that all its counterparts in worlds that contain at least one G thing are F.

Universal closures of instances of K (and CC) would however become theorems if we added A6.

How does this play out in English?

12. 'Counterparts of persons and their bodies'

13. Fine

Fine is interested in monists who in some cases accept 'The statue is F' and 'The [piece of] alloy isn't F' but also accept 'The statue is identical to the alloy'.

- They could diagnose this in terms of context-shift, or allow all three to be true in the same context; this contrast doesn't play a role in Fine's discussion.

Reconstruction of Fine's argument that such monists are committed to "predicational shift".

(1) Fully quantified Leibniz's Law: $\forall Z \forall x \forall y(x=y \rightarrow (Zx \rightarrow Zy))$. Identity implies having the same properties.

(2) If there is a property Z such that 'well-made' refers to Z in both 'The statue is well-made' and 'The alloy is not well-made', then whatever 'the statue' refers to in the former is distinct from whatever 'the alloy' refers to in the latter.

(3) Whatever 'the statue' refers to in 'The statue is well-made' coincides with whatever 'the alloy' refers to in 'The alloy is not well-made'.

(4) Therefore, if monism is true, there is no property Z such that 'well-made' refers to Z in both 'The statue is well-made' and 'The alloy is not well-made'.

§5: Evidence that predicational shift is driven by a "semantic mechanism" (as opposed to something more flexible involving different readings being "naturally suggested" in different linguistic surroundings). Claim: no way of filling in the dots in 'The alloy that was shaped… is well-made' will make it mean what 'The statue is well made' means.

In §6, Fine imagines an elaborate specification of the semantic mechanism that can generalise to get 'The item Al created is well-made' the intuitive truth conditions:

> We may suppose that certain singular terms are assigned both a referent and a sort and that certain predicate-expressions are only assigned an extension relative to a sort (or perhaps several sorts, one for each of several arguments). There are then systematic rules for determining how these sorts play out in the semantic evaluation of more complex expressions. Suppose, for example, that $\phi(x)$ is a sort-relative predicate-expression. Then when t is a term to which has been assigned the referent r and the sort s, $\phi(t)$ will be true iff $\phi(x)$ is true of r relative to s (that is, $\phi(x)$ is true under the assignment of r and s as referent and sort to the variable x). Or the referent r and the sort s may be assigned to the definite description 'the x such that $\phi(x)$' just in case (r, s) is the unique object-sort pair for which $\phi(x)$ is true of r relative to s. Or again, 'some x is such that $\phi(x)$' will be true iff $\phi(x)$ is true of some object r relative to a sort s. (220)